🔑 **Answer Key — Teacher Copy · Lesson 4-2**

MSTAR PRACTICE WORKSHEET

Lesson 4-2: Relate Fractions, Decimals, and Percentages

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

Convert 3/5 to a percent.

A. 35%

B. 53%

C. 60%

D. 65%

5 × 20 = 100, so 3 × 20 = 60. That makes 3/5 = 60/100 = 0.6 = 60%.

Question 2 SELECTED RESPONSE — CORRECT: C

Convert 7/20 to a percent.

A. 14%

B. 20%

C. 35%

D. 70%

20 × 5 = 100, so 7 × 5 = 35. That makes 7/20 = 35/100 = 0.35 = 35%.

Question 3 SELECTED RESPONSE — CORRECT: B

Which decimal is equivalent to 45%?

A 0.045

B 0.45

C 4.5

D 45.0

45% means 45/100, and 45/100 written as a decimal is 0.45 (forty-five hundredths).

Question 4 SELECTED RESPONSE — CORRECT: A

What fraction is equal to 0.75?

A 3/4

B 3/5

C 7/5

D 7/10

0.75 = 75/100. Simplify by dividing both by 25: 75÷25 = 3 and 100÷25 = 4. So 0.75 = 3/4.

Question 5 SELECTED RESPONSE — CORRECT: C

A game shows your score as 7/8 complete. What percent is that?

A 75%

B 78%

C 87.5%

D 88%

8 does not scale to 100 by a whole number, so divide: 7 ÷ 8 = 0.875, and 0.875 × 100 = 87.5%. That is 87.5 out of 100.

Question 6 SELECTED RESPONSE — CORRECT: B

Your arcade scoreboard reads 0.25 of the level cleared. What percent and fraction match this score?

A. 2.5% and 1/4

B. 25% and 1/4

C. 25% and 1/5

D. 250% and 1/40

Move the decimal two places: 0.25 = 25%. As a fraction, 0.25 = 25/100 = 1/4.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

What is the fraction 7/20 written as a decimal and as a percent?

A. 0.035 and 3.5%

B. 0.35 and 35%

C. 0.35 and 3.5%

D. 0.7 and 70%

7/20 = (7 × 5) / (20 × 5) = 35/100 = 0.35 = 35%.

- **A:** The decimal point is one place too far left. 7 divided by 20 is 0.35.
- **C:** The decimal is right but the percent is not. 0.35 is 35 hundredths, so it is 35%.
- **D:** That is 7/10, not 7/20. Scale to hundredths: 7/20 equals 35/100.

Part B — correct answer: A

Which mathematical reasoning justifies the conversion in Part A?

A. Multiplying both numerator and denominator by 5 yields the equivalent fraction 35/100, which equals 0.35 and 35%.

B. Moving the decimal point in 7.20 one place left gives 0.72.

C. Adding 15 to both terms gives 22/35, which equals 35%.

D. Dividing 20 by 7 gives approximately 2.85, so the percent is 35%.

Scaling the denominator to 100 by multiplying by 5/5 preserves equivalence and directly reveals the rate per hundred.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Spot the Decimal-Percent Error

> Elena wrote that 0.08 is equal to 80% because 'there is an 8 in the number.'

Explain Elena's misconception and determine the correct percent value.

Model answer: 0.08 = 8%. Elena confused hundredths with tenths; 80% is 0.80 (80/100), whereas 0.08 is 8 hundredths (8/100 = 8%).

2 points	Student clearly identifies that 0.08 has 8 in the hundredths place (8/100 = 8%) and explains that 80% is 0.80 (80/100).
1 point	Student provides the correct answer (8%) but provides an incomplete or vague explanation of place value.
0 points	Student agrees with Elena or gives an incorrect response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student says 3/8 is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

Model answer: Bigger-looking digits do not mean a bigger value — the two amounts have to be in the same form before you compare them. Converting the fraction gives 3 ÷ 8 = 0.375, which is 37.5%. Since 37.5% is less than 40%, the student is not correct: 3/8 is the smaller value.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.